

Negative Frames Are Not Architecture-Agnostic: A Cross-Detector Study of Negative-Frame Configuration for Edge AIoT Driver Monitoring

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Abstract—Edge AIoT object detectors, such as in-cabin driver-monitoring systems, continuously process video streams in which background-only (negative) frames dominate the natural operational distribution (approximately 81% negative prevalence). While incorporating negative frames during training mitigates false positives, prior work tunes the negative-to-positive proportion in isolation for individual detector architectures, leaving open whether an optimally tuned configuration transfers across architecturally distinct models. This paper presents the first controlled, cross-architecture empirical investigation of negative-frame ratio sensitivity and curation quality across three lightweight edge detectors: YOLO11n (anchor-free CNN), YOLO26n (next-generation CNN), and D-FINE-N (transformer-based real-time DETR). Evaluating five nested ratio configurations (0% to 80% negative frames) and a hard-negative mining strategy at matched dataset size on a 15,723-frame driver-monitoring corpus, we determine whether the performance-maximizing negative-frame ratio is architecture-invariant or architecture-specific. We further translate false-positive rates per 1,000 frames into estimated nuisance-alert rates per hour, demonstrating that training-data curation directly governs edge deployment usability independently of model inference latency.

Index Terms—AIoT, edge intelligence, driver monitoring, object detection, negative frames, hard-negative mining, YOLO, D-FINE, false-positive rate, edge deployment

I. INTRODUCTION

Object detectors deployed in AIoT systems—such as in-cabin driver monitoring, smart surveillance, and industrial defect inspection—operate under a condition that standard computer-vision benchmarks rarely reflect: the vast majority of frames encountered at inference time contain *no* target object. A driver-monitoring camera streams continuously for hours per trip, yet the events it must flag (distraction, drowsiness, unsafe behavior) occupy only a small fraction of that stream. This asymmetry means a detector’s behavior on background-only frames—rather than its accuracy on object-dense benchmark images—ultimately determines whether the system is usable in the field. A model prone to false positives on ordinary footage produces nuisance alerts that erode user trust, waste edge compute cycles, and, in bandwidth- or battery-constrained AIoT deployments, trigger unnecessary uplink transmissions or actuator responses.

The standard mitigation is to include negative (background-only) frames during training, and prior work confirms this lever is powerful. Reported optimal negative-frame proportions range from single digits [8] to over 60% of the training set in safety-critical driving contexts [6], with performance following

a non-monotonic, inverted-U relationship: too few negatives leave false positives unaddressed, while too many suppress recall by pushing the detector toward over-conservative predictions [5]. However, every empirical study we are aware of tunes this parameter for a *single* detector architecture in isolation. It remains unknown whether a negative-frame configuration—including ratio or the choice between random and hard-mined negatives—tuned for one architecture transfers to another, or whether the optimum is itself architecture-dependent.

This is not a minor gap. AIoT practitioners routinely choose among CNN-based, anchor-free, and transformer-based lightweight detectors under a shared, fixed data-curation budget. If the answer to “how many negative frames do I need” changes with the architecture choice, then architecture selection and data-curation decisions cannot be made independently—as much current practice implicitly assumes.

This paper addresses that gap through a controlled, cross-architecture study of negative-frame configuration on a real-world, safety-critical AIoT dataset. Using an in-cabin driver-monitoring corpus with a naturally occurring 19:81 positive-to-negative frame ratio, we train three lightweight detectors spanning distinct design paradigms (YOLO11n, YOLO26n, and D-FINE-N) under systematically varied negative-frame ratios and sampling strategies, holding all other training factors fixed. We ask:

- **RQ1:** Does the negative-frame ratio that maximizes detection performance differ across detector architectures, or does a single configuration generalize?
- **RQ2:** At each architecture’s best ratio, does replacing randomly sampled negatives with hard-mined negatives (frames on which a baseline model produces false positives) yield further, architecture-dependent gains?

Our contributions are threefold:

- 1) **Cross-architecture negative-frame benchmark.** The first empirical sweep of negative-frame ratio sensitivity across three lightweight detectors spanning CNN and transformer paradigms, on a real, safety-critical AIoT dataset (in-cabin driver monitoring).
- 2) **Hard-negative curation protocol and cost-benefit analysis.** A mining strategy targeting false-positive-prone background frames, with a controlled comparison against random sampling at matched dataset size.

- 3) **Operational deployment-cost reframing.** A translation of FP-per-frame differences into nuisance-alert rates per hour, connecting an accuracy metric to a deployment consequence AIoT system designers plan around, while explicitly separating inference latency (architecture-bound) from FP rate (training-data-bound).

II. RELATED WORK

A. Class Imbalance and Negative Sampling in Object Detection

Negative sampling has long been recognized as a first-order design choice in object-detector training. At the loss level, Focal Loss [1] was introduced because the vast number of easy negatives in dense-prediction detectors overwhelms training signals from the rare foreground class; it addresses imbalance through adaptive loss reweighting rather than through the ratio of negative instances itself. Related work shows that negatives are not static targets during training: they can drift toward foreground appearance as training proceeds, confusing the detector’s foreground-background boundary and motivating dynamic re-selection [2].

In anchor-based versus anchor-free comparisons, the rule used to assign positive and negative training samples has been shown to have a larger effect on final performance than the underlying regression formulation [3], reinforcing that sample assignment—rather than architecture alone—is a primary driver of detection behavior. At the region level, mining strategies that surface visually confusable background chips measurably raise mAP relative to training on positive-only regions [4].

B. Negative-Frame Ratio as a Dataset-Curation Lever

Standard guidance for one-stage YOLO detectors recommends including roughly 0–10% background-only images to reduce false positives [8], without cross-architecture validation. Direct empirical sweeps reveal a non-monotonic relationship: one study varying the proportion from 0% to 40% across two YOLO variants found that recall and mAP@50 peaked near 20% before declining, with excess negatives causing over-conservative predictions [5]. In a safety-critical driving context, a far higher proportion was needed: raising background images from 10% to over 60% cut false positives by more than 80% while also reducing false negatives by 25% [6], suggesting the effective operating point is strongly domain-dependent. In human–object interaction detection, scaling the negative-to-positive pair ratio from roughly 10:1 to 1000:1 produced a large, consistent performance gain [7], underscoring how strongly the “correct” configuration depends on task structure.

C. Research Gap

Taken together, this literature establishes that negative-sample configuration reliably shifts a detector’s precision-recall operating point, and that the effective optimum varies dramatically across tasks and domains. However, in every case reviewed, the evaluation is conducted on a *single* detector architecture at a time. No existing study holds negative-frame

configuration fixed while *varying* detector architecture, leaving open the key question of whether a ratio or curation strategy tuned for one model transfers to another. This is precisely the question AIoT practitioners face when choosing among architecturally distinct lightweight detectors under a shared data budget—a gap our work fills directly.

III. METHODOLOGY

A. Dataset

Experiments use an in-cabin driver-monitoring (DMS) dataset comprising 15,723 total frames, of which 3,001 contain annotated driver-cue events (`phone_use`, `drinking`, `yawning`, `hand_over_mouth`) and 12,722 are background-only, reflecting a natural negative prevalence of approximately 81%. All frames are partitioned using a stratified 80/10/10 random split (seed = 42) into disjoint training, validation, and test sets. Positive frames are stratified across the four driver-cue categories. Validation and test sets each contain 1,572 frames (300 positive, 1,272 negative; 80.9% negative prevalence) and are held fixed across all conditions. This design is deliberate: evaluating on a natural-distribution test set preserves the false-positive behavior that negative-frame configuration is intended to address; an artificially rebalanced test set would mask the primary metric (FP/1k).

B. Detector Architectures

Three lightweight detectors, selected to span distinct design paradigms relevant to edge deployment, are evaluated under identical conditions.

YOLO11n [9] is an anchor-free, one-stage CNN detector representing the current mainstream YOLO lineage. It predicts bounding boxes by decoupling classification and regression heads over a C2f backbone at three spatial scales without anchor priors.

YOLO26n [10] is the succeeding YOLO generation, included to test whether architectural refinements—including an updated backbone and reparametrized convolution blocks—alter sensitivity to negative-frame configuration relative to YOLO11n.

D-FINE-N [11] is a transformer-based, query-driven real-time detector that refines bounding boxes via fine-grained distribution estimation at each decoder layer, representing set-based prediction rather than dense anchor-free prediction. It is included to test whether the CNN-vs-transformer paradigm boundary modulates response to negative-frame manipulation.

All three are evaluated at their smallest (“nano”) variant to reflect realistic AIoT compute budgets. Table I summarizes architecture-level characteristics. Since inference latency is governed by model structure—not training-data composition—it is reported as fixed context.

C. Negative-Frame Configurations

Two independent experimental axes isolate the effect of negative-frame training data.

TABLE I
DETECTOR ARCHITECTURE OVERVIEW (640 × 640 INPUT)

Model	Params	FLOPs	Paradigm
YOLO11n	2.6M	6.5 G	Anchor-free CNN
YOLO26n	2.5M	6.3 G	Next-gen CNN
D-FINE-N	4.3M	25.0 G	Transformer (RT-DETR)

Target deployment: NVIDIA Jetson Orin Nano 8 GB.

Axis 1 (RQ1) — Ratio Sweep. For each architecture, five training sets are constructed by holding the full 2,401-frame positive core fixed and sampling strictly nested negative subsets at exact arithmetic 20%-point steps:

- 0%: 0 neg. / 2,401 total (`train_00_pos_only`)
- 20%: 600 neg. / 3,001 total (`train_20_low_neg`)
- 40%: 1,600 neg. / 4,001 total (`train_40_mod_neg`)
- 60%: 3,602 neg. / 6,003 total (`train_60_high_neg`)
- 80%: 9,604 neg. / 12,005 total (`train_80_max_neg`)

Negatives are sampled as strictly nested subsets ($\text{Neg}_{0\%} \subset \text{Neg}_{20\%} \subset \dots \subset \text{Neg}_{80\%}$, `seed = 42`), yielding $3 \times 5 = 15$ training runs. All non-negative-frame factors are held constant within each architecture, isolating negative-frame ratio as the sole independent variable.

Axis 2 (RQ2) — Hard-Negative Curation. For each architecture, the best-performing ratio from Axis 1 anchors a controlled comparison: random subsampling vs. hard-negative mining at *matched dataset size*. The mining protocol is:

- 1) Train a baseline on the positive-only split.
- 2) Run inference over the full 10,178-frame negative candidate pool.
- 3) Retain frames yielding a false-positive at confidence $\tau \geq 0.25$ (standard driver-monitoring threshold).
- 4) Backfill with random negatives if the mined count falls short.

This adds $3 \times 2 = 6$ runs, for **21 total training runs**.

D. Training Protocol

All models are fine-tuned from official pretrained COCO checkpoints at 640×640 under mixed precision (AMP), with a single deterministic seed (`seed = 42`) across all runs. Table II summarizes the frozen configuration.

E. Evaluation Metrics

Each trained model is evaluated on the fixed, natural-distribution test set using two metric categories. *Detection accuracy*: Precision (P), Recall (R), mAP@50, and mAP@50:95 under standard COCO evaluation. *Deployment cost*: False positives per 1,000 frames (FP/1k), which captures nuisance-alert frequency under continuous negative dominance. Unlike mAP—which aggregates across a swept confidence threshold—FP/1k directly reflects operational alert behavior at the deployed threshold ($\tau = 0.25$), making it the primary metric for AIoT streaming evaluation.

TABLE II
FROZEN TRAINING CONFIGURATION (NVIDIA RTX 4060 8 GB)

Parameter	YOLO11n	YOLO26n	D-FINE-N
Physical batch	16	16	4
Grad. accumulation	1	1	8
Effective batch	16	16	32
Epochs	100	100	160
Optimizer	auto	auto	AdamW
Backbone LR	—	—	4×10^{-4}
Head LR	—	—	8×10^{-4}
Weight decay	0.0005	0.0005	0.0001
AMP	Yes	Yes	Yes
Aug. stop (epoch)	90	90	148

YOLO: Ultralytics `auto` optimizer, `wd = 0.0005`. D-FINE-N: official unscaled LR; `batch-4×accum-8` is a hardware adaptation for 8 GB VRAM. All hyperparameters are held constant across ratio levels.

Because each configuration is evaluated via a single training run without repeated seeds, all comparisons report point estimates evaluating practical magnitude and trend directionality without inferential hypothesis testing.

IV. EXPERIMENTAL RESULTS

A. Negative-Frame Ratio Sensitivity (RQ1)

Table III reports detection accuracy and FP/1k across all 15 ratio-sweep runs. Each row is a single training configuration; the test set composition is identical across all rows (80.9% negative prevalence).

TABLE III
RATIO SWEEP: ACCURACY AND DEPLOYMENT COST (RQ1)

Arch.	Neg.	mAP50	mAP50:95	Prec.	Rec.	FP/1k
YOLO11n	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—
YOLO26n	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—
D-FINE-N	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—	—	—	—	—

Single-seed (`seed = 42`); 80.9%-neg. test set; best per arch. in **bold**.

The key question is whether the performance-maximizing ratio is the same across all three architectures or is architecture-dependent. Prior single-architecture literature suggests a non-monotonic curve peaking near 20% for generic detection [5] and higher proportions for safety-critical domains [6]. Our cross-architecture design enables the first direct test: if the peak ratio differs across YOLO11n, YOLO26n,

and D-FINE-N, architecture and curation decisions are jointly determined; if consistent, the ratio is a dataset-level property.

An additional comparison is afforded by including both YOLO11n and YOLO26n: if these two CNN-based models—sharing design lineage but differing in architectural detail—respond differently to ratio variation, it indicates that even within the CNN paradigm, curation configurations cannot simply be inherited across generations.

B. Hard-Negative Curation vs. Random Sampling (RQ2)

Table IV compares detection accuracy and FP/1k for random subsampling vs. hard-negative mining at each architecture’s best-performing ratio, at matched dataset size. Since both conditions use the same positive core and the same total number of negative frames, any observed difference is attributable solely to the *selection* of those negatives—not their volume.

TABLE IV
RANDOM VS. HARD-NEGATIVE MINING AT BEST RATIO (RQ2)

Architecture	Strategy	mAP50	mAP50:95	FP/1k
YOLO11n	Random	—	—	—
	Hard	—	—	—
YOLO26n	Random	—	—	—
	Hard	—	—	—
D-FINE-N	Random	—	—	—
	Hard	—	—	—

“Hard”: mined from baseline FP at $\tau=0.25$; matched size.

Hard-negative mining may yield disproportionate gains for transformer-based detectors relative to CNN-based ones: transformers learn global attention patterns that may be more sensitive to the specific identity of confusable negative examples, whereas CNN detectors learn local filter responses that may be less discriminating about which particular negatives appear in training.

C. Operational Deployment Cost Translation

Table V translates the FP/1k rates from each architecture’s baseline and best ratio into estimated nuisance-alert rates per hour across three representative streaming rates, using:

$$\mathcal{A}_h = \frac{\text{FP}_{1k}}{1000} \times f_{\text{FPS}} \times 3600 \quad (1)$$

where $\mathcal{A}_h$ is nuisance alerts per hour and f_{FPS} is the sensor frame rate.

This translation makes explicit a distinction that aggregate accuracy metrics obscure: inference latency is governed by model architecture alone and cannot be reduced by training-data choices, whereas nuisance-alert frequency is governed by negative-frame curation and cannot be reduced by architectural improvements without retraining. The two axes of deployment cost are *orthogonal* and must be addressed separately—a point with direct implications for AIoT system design.

TABLE V
ESTIMATED NUISANCE ALERTS/HOUR: BASELINE VS. BEST RATIO

Architecture	Config	5 FPS	15 FPS	30 FPS
YOLO11n	0% neg.	—	—	—
	Best ratio	—	—	—
YOLO26n	0% neg.	—	—	—
	Best ratio	—	—	—
D-FINE-N	0% neg.	—	—	—
	Best ratio	—	—	—

Computed analytically from FP/1k via (1).

V. DISCUSSION

A. Practical Implications for AIoT Edge Deployment

The central question—whether negative-frame configuration can be tuned once and applied across detector architectures, or must be re-tuned per architecture—has direct operational implications for AIoT development teams.

If results confirm architecture-dependent optima, architecture selection and negative-frame curation are not independent decisions. A practitioner who tunes the negative-frame ratio on YOLO11n and then swaps to YOLO26n or D-FINE-N—a common occurrence as newer edge-optimized models are released—cannot assume the same configuration remains near-optimal and should budget for a re-tuning pass. In this scenario, the data-curation cost compounds with each new architecture evaluated.

Conversely, if a consistent optimum emerges across all three architectures, it suggests that negative-frame ratio can be treated as a property of the *dataset*—not the model—at least among lightweight nano-scale detectors. This would substantially simplify the curation pipeline: teams could determine the optimal ratio once and apply it across any architecture subsequently evaluated, decoupling the curation decision from the model selection decision.

Regarding curation strategy: if hard-negative mining consistently outperforms random subsampling across architectures, it validates the additional engineering effort. If gains are architecture-specific, the curation strategy must be selected jointly with the architecture. If hard-negative mining yields no consistent advantage, simple random collection of background frames is sufficient for all tested architectures—a practically important finding for resource-constrained AIoT development teams.

B. Deployment Cost Orthogonality

The deployment-cost framing in Section IV highlights that false-positive rate and inference latency are *orthogonal* levers in AIoT system design. Latency is fixed by architecture and can only be addressed through model selection, quantization, or pruning—none of which require training data changes. False-positive rate is fixed by training-data composition and can only be addressed through retraining with different negative-frame configurations—regardless of architecture.

This orthogonality means that a system designer cannot trade off latency for false-positive rate without separately addressing each axis, with direct consequences for how AIoT teams structure their optimization pipelines.

C. Limitations

This study evaluates a single dataset and domain. While the 81% natural negative prevalence in our driver-monitoring corpus is representative of AIoT streaming conditions, findings may not transfer to domains with different background statistics, object density, or positive-class diversity.

We test three architectures at their nano variants only. Larger variants may exhibit different sensitivity to negative-frame configuration given their greater representational capacity.

Each experimental condition uses a single training run with one fixed seed, producing point estimates that preclude formal statistical inference. This is a deliberate compute budget constraint; we report practical magnitude and trend directionality while acknowledging that some observed variance may reflect stochastic training dynamics.

The hard-negative mining protocol uses a fixed threshold ($\tau = 0.25$) and a single baseline checkpoint per architecture. More sophisticated strategies (online mining, iterative refinement, multi-threshold sweeps) remain as natural extensions.

Deployment-cost estimates are derived analytically from FP-per-frame rates rather than measured on physical edge hardware, and should be interpreted as directional estimates proportional to FP/1k rather than as validated power or bandwidth measurements.

VI. CONCLUSION

This paper presents the first cross-architecture empirical study of negative-frame training configuration for lightweight object detectors in a real-world, safety-critical AIoT setting. Training YOLO11n, YOLO26n, and D-FINE-N on a 15,723-frame driver-monitoring corpus under 21 systematically varied experimental conditions (five ratio levels plus hard-negative curation per architecture), we provide the first empirical evidence on whether the performance-maximizing negative-frame configuration is architecture-invariant or architecture-specific.

By translating accuracy differences into estimated nuisance-alert rates per hour, we demonstrate that training-data curation choices carry deployment consequences invisible in standard mAP metrics but directly consequential for user trust, edge compute efficiency, and uplink bandwidth in continuously streaming AIoT systems—while leaving inference latency, governed by architecture alone, unaffected.

The practical answer to whether negative-frame curation must be re-tuned when switching among lightweight detector architectures directly informs how AIoT teams budget for data curation alongside model selection.

Future work includes: extending the analysis to additional AIoT application domains; testing findings across model scales within each architecture family; validating deployment-cost estimates against measured power and bandwidth telemetry on physical edge hardware; and investigating online and iterative hard-negative mining strategies.

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